NeST-Com: Fixed-Budget Next-Scale Token Communication

Supplementary Configurations and Results

S1. Configuration and calibration

This supplement collects the completed experiments supporting the accompanying letter. It reformats existing results; it adds no model inference, wireless trial, metric evaluation, training, or statistical resampling. All differences below retain the original comparison direction and population.

Models and training. NeST and WHOLE share the VAR tokenizer, null-class prior, entropy coder, and adapted image decoder D_c. Its adaptation uses 20 000 ImageNet training images and continuous, fully quantized, and prefix-completion-interpolated latents. The convolutional Latent JSCC uses MSE, LPIPS, and latent-consistency losses and the same decoder; its final checkpoint is at 100k updates. SwinJSCC uses the same 20 000 images, 256×256 input, paid side information, and a final 200k-update checkpoint. The Swin selection record reports `BUDGET_CAP_WHILE_IMPROVING`; no claim of full convergence is made. The selected checkpoint, calibration history, stopping checks, and original training plots are available through the linked index.

Ada adaptation. Ada retains LlamaGen-L/VQ16 and genuine autoregressive prefix arithmetic coding. A fixed ConvNeXt predicts the class at TX; its 10-bit ID travels in the protected body, giving $b + 39 \leq k_a$. RX uses the recovered class and prefix. All 1–256 independently terminated prefixes are eligible. Main-curve tail generation uses CFG=4, temperature 1, top-$k = 2000$, top-$p = 1$, FP32 with TF32 off. The fixed single-frame adaptation does not reproduce the original long-term average-budget controller or ARQ. At N1024/19 dB all 256 tokens are sent; remaining source capacity therefore does not indicate fitting failure.

Calibration. The 100-image pilot uses one channel realization per source. Final evaluation uses all 1000 calibration images with three realizations per source, including fixed-gray failures. Selection maximizes DINOv2 cosine, first averaging realizations and then equally weighting sources. Exact ties use the frozen candidate order. DINO-based selection is related to feature-based evaluation; the controlled interventions separately examine the mechanism.

N1024 target screening starts at scales 7–9 and includes full-scale $(10, 0)$ candidates in final evaluation. N512 screens all 7 WHOLE and 25 Quarter targets against 12 MCSs: $\log_2 M \in \{2, 4, 6\}$ and rates $\{1/2, 2/3, 3/4, 5/6\}$. Each method/SNR advances the top three and a maximum-endpoint QPSK rate-1/2 anchor. Ada uses the same N512 MCS pool and up to 256 tokens. Each budget is independently calibrated; its results describe separately selected systems, rather than an isolated causal change in budget or representation order.

Swin checkpoint and training status; training and calibration curves; N512 frozen protocol.

S2. Statistical populations and reception accounting

The main population is common500; Kodak has 24 sources. The endpoint, receiver, and E1/E2 controls use the existing fixed 100-source subset, not a new independent blind test. Full source IDs, preprocessing identities, and native pixel hashes join the records. Three original outcomes are averaged within a source before equal-source means and paired differences. Percentile 95% intervals use 10 000 source resamples; they are pointwise, not simultaneous intervals. Marginal intervals are never subtracted.

N512 methods share the registered source/repeat AWGN identities. N1024 preserves historical noise identities and is not presented as sharing noise across every method or budget. E1/E2 reuse those actual reception states; receiver counterfactuals are not new wireless frames. The 100-source bank is matched by its complete source order, not sliced from the 500-source bank. Both registered banks use seed 2026100701. Full hashes and source orders remain in the statistical identity files.

Main matched-SNR failures. N1024 has no rejected or failed reconstructions in 1500 trials per method/SNR; continuous Latent JSCC has no discrete CRC rejection. N512 failures below are body-CRC rejections, retained as fixed gray in all means. Header, parser, and source-decoder rejection counts are zero. No evaluated N512 source requires a NULL frame.

Method	4 dB	10 dB	19 dB	Trials per SNR
NeST	4	1	0	1500
WHOLE	7	0	0	1500
Ada	5	0	0	1500

No-feasible-endpoint behavior. The actual implementations differ. N512 sends a paid NULL profile and two zero payload bits through the complete 68+444-symbol frame, returning gray. N1024 raises an execution error when no independently terminated legal stream fits; this case did not occur in its evaluated sources. The revision preserves both frozen implementations. Seven synthetic CPU checks of the original source-selection/RX-projection functions verify no-fit and ordinary-fit branches, with no PHY or model calls. Source hashes and line ranges are in the implementation audit; these checks do not constitute new wireless qualification.

Granularity and receiver controls. Mean terminated-length checks are 3.55/5.55/9.49/17.58 for Whole/Half/Quarter/Eighth. They compare already terminated streams, not extra model forwards or measured latency. Quarter is a predefined configuration, not a demonstrated optimum. The A0/B0/C0 differences are ordered interventions on the same prefixes; they are not a universal percentage decomposition. A0 still performs prior completion. At the final scale C0 and B0 coincide because no finer context is used.

S3. Ada tail sensitivity: $N = 512$, 4 dB

All entries use the same 100 sources and three original repetitions (300 logical results per arm). Each cell is a mean or direct paired difference with its 95% source-bootstrap interval. Higher PSNR/CLIP-I and lower LPIPS/DreamSim are favorable.

Absolute metrics

Metric	Ada sampling	Ada argmax	NeST
PSNR (dB) ↑	10.984 [10.574, 11.410]	10.509 [10.099, 10.933]	15.141 [14.621, 15.673]
LPIPS ↓	0.5610 [0.5457, 0.5759]	0.6033 [0.5857, 0.6209]	0.3668 [0.3511, 0.3823]
DreamSim ↓	0.3797 [0.3523, 0.4071]	0.4247 [0.3966, 0.4534]	0.2786 [0.2591, 0.2981]
CLIP-I ↑	0.7922 [0.7740, 0.8095]	0.7661 [0.7466, 0.7851]	0.7658 [0.7468, 0.7855]

Direct source-paired differences

Metric	Argmax minus sampling	NeST minus argmax	NeST minus sampling
PSNR (dB) ↑	-0.475 [-0.801, -0.152]	4.632 [4.183, 5.093]	4.157 [3.772, 4.558]
LPIPS ↓	0.0423 [0.0294, 0.0555]	-0.2365 [-0.2567, -0.2167]	-0.1942 [-0.2100, -0.1789]
DreamSim ↓	0.0450 [0.0282, 0.0621]	-0.1461 [-0.1709, -0.1218]	-0.1010 [-0.1248, -0.0769]
CLIP-I ↑	-0.0261 [-0.0378, -0.0143]	-0.0003 [-0.0194, 0.0185]	-0.0263 [-0.0449, -0.0078]

Fixed-gray counts per 300: Ada-sampling: 0, Ada-argmax: 0, NeST: 1. The deterministic tail lowers Ada PSNR and worsens its three feature metrics relative to sampling. NeST retains higher PSNR and lower LPIPS/DreamSim than argmax; the CLIP-I difference interval crosses zero. The sampling configuration remains the main baseline. Only unreceived-tail token selection changes. Context/KV is rebuilt from the actual received prefix and follows each new greedy token. The paid class, CFG composition, entropy-decoding conditional CDF, model, decoder, and numerical route remain fixed. No sampling-tail KV is reused. Full-token receptions reuse the existing output exactly; failed/NULL inputs remain gray.

S3. Ada tail sensitivity: $N = 512$, 10 dB

All entries use the same 100 sources and three original repetitions (300 logical results per arm). Each cell is a mean or direct paired difference with its 95% source-bootstrap interval. Higher PSNR/CLIP-I and lower LPIPS/DreamSim are favorable.

Absolute metrics

Metric	Ada sampling	Ada argmax	NeST
PSNR (dB) ↑	12.878 [12.384, 13.398]	12.227 [11.738, 12.751]	16.827 [16.313, 17.347]
LPIPS ↓	0.4121 [0.3975, 0.4266]	0.4416 [0.4237, 0.4598]	0.2818 [0.2683, 0.2955]
DreamSim ↓	0.2511 [0.2296, 0.2728]	0.2856 [0.2627, 0.3098]	0.1955 [0.1797, 0.2123]
CLIP-I ↑	0.8337 [0.8170, 0.8502]	0.8105 [0.7929, 0.8278]	0.8208 [0.8028, 0.8386]

Direct source-paired differences

Metric	Argmax minus sampling	NeST minus argmax	NeST minus sampling
PSNR (dB) ↑	-0.651 [-0.984, -0.338]	4.600 [4.188, 5.018]	3.949 [3.566, 4.348]
LPIPS ↓	0.0296 [0.0201, 0.0394]	-0.1598 [-0.1754, -0.1439]	-0.1302 [-0.1429, -0.1178]
DreamSim ↓	0.0345 [0.0232, 0.0461]	-0.0901 [-0.1120, -0.0689]	-0.0556 [-0.0760, -0.0360]
CLIP-I ↑	-0.0231 [-0.0337, -0.0128]	0.0102 [-0.0072, 0.0280]	-0.0129 [-0.0300, 0.0039]

Fixed-gray counts per 300: Ada-sampling: 0, Ada-argmax: 0, NeST: 1. The deterministic tail lowers Ada PSNR and worsens its three feature metrics relative to sampling. NeST retains higher PSNR and lower LPIPS/DreamSim than argmax; the CLIP-I difference interval crosses zero. The sampling configuration remains the main baseline. Only unreceived-tail token selection changes. Context/KV is rebuilt from the actual received prefix and follows each new greedy token. The paid class, CFG composition, entropy-decoding conditional CDF, model, decoder, and numerical route remain fixed. No sampling-tail KV is reused. Full-token receptions reuse the existing output exactly; failed/NULL inputs remain gray.

S3. Ada tail sensitivity: $N = 1024$, 10 dB

All entries use the same 100 sources and three original repetitions (300 logical results per arm). Each cell is a mean or direct paired difference with its 95% source-bootstrap interval. Higher PSNR/CLIP-I and lower LPIPS/DreamSim are favorable.

Absolute metrics

Metric	Ada sampling	Ada argmax	NeST
PSNR (dB) ↑	19.200 [18.592, 19.824]	19.044 [18.403, 19.698]	19.612 [19.083, 20.146]
LPIPS ↓	0.1717 [0.1596, 0.1845]	0.1755 [0.1625, 0.1890]	0.1752 [0.1653, 0.1851]
DreamSim ↓	0.0863 [0.0795, 0.0933]	0.0899 [0.0823, 0.0978]	0.1077 [0.1000, 0.1157]
CLIP-I ↑	0.8897 [0.8764, 0.9023]	0.8884 [0.8751, 0.9008]	0.8759 [0.8607, 0.8905]

Direct source-paired differences

Metric	Argmax minus sampling	NeST minus argmax	NeST minus sampling
PSNR (dB) ↑	-0.156 [-0.369, 0.057]	0.568 [0.246, 0.902]	0.412 [0.157, 0.677]
LPIPS ↓	0.0038 [0.0007, 0.0071]	-0.0003 [-0.0072, 0.0062]	0.0034 [-0.0021, 0.0089]
DreamSim ↓	0.0036 [0.0010, 0.0060]	0.0178 [0.0129, 0.0229]	0.0214 [0.0167, 0.0262]
CLIP-I ↑	-0.0013 [-0.0052, 0.0025]	-0.0124 [-0.0222, -0.0025]	-0.0137 [-0.0235, -0.0038]

Fixed-gray counts per 300: Ada-sampling: 0, Ada-argmax: 0, NeST: 0. The argmax PSNR difference from sampling includes zero, while DreamSim becomes worse. NeST retains higher PSNR than argmax; Ada argmax retains better DreamSim and CLIP-I, and the LPIPS interval crosses zero. Decoding sensitivity does not remove other complete-system differences. Only unreceived-tail token selection changes. Context/KV is rebuilt from the actual received prefix and follows each new greedy token. The paid class, CFG composition, entropy-decoding conditional CDF, model, decoder, and numerical route remain fixed. No sampling-tail KV is reused. Full-token receptions reuse the existing output exactly; failed/NULL inputs remain gray.

S4. Completion versus direct coarse decoding

R0 decodes $D_c(\sum_{s=1}^{m} C_s(q_s))$ from the actual received WHOLE prefix. R1 is the existing VAR-completed reconstruction of that same prefix. Both retain the original ten-scale indexing, scale-dependent residual transforms, full latent grid, frozen adapted D_c, clipping, and metric preprocessing. Unreceived scales contribute nothing to R0; codebook ID zero is not used as zero contribution. Neither arm uses unsent source tokens or a continuous source latent. This is distinct from A0/B0/C0, which intervene on received boundary-token use.

$N = 512$, 4 dB: 100 sources, 300 results per arm

Metric	R0 coarse	R1 completed	R1 minus R0
PSNR (dB) ↑	14.787 [14.293, 15.296]	14.813 [14.261, 15.383]	0.026 [-0.204, 0.271]
LPIPS ↓	0.6612 [0.6376, 0.6843]	0.3960 [0.3752, 0.4172]	-0.2652 [-0.2806, -0.2500]
DreamSim ↓	0.7369 [0.7154, 0.7575]	0.3166 [0.2915, 0.3421]	-0.4204 [-0.4488, -0.3925]
CLIP-I ↑	0.5493 [0.5337, 0.5649]	0.7521 [0.7303, 0.7737]	0.2028 [0.1837, 0.2218]

Fixed-gray count per arm: 3.

$N = 512$, 10 dB: 100 sources, 300 results per arm

Metric	R0 coarse	R1 completed	R1 minus R0
PSNR (dB) ↑	15.722 [15.235, 16.209]	16.493 [15.993, 17.004]	0.771 [0.563, 0.986]
LPIPS ↓	0.5581 [0.5388, 0.5771]	0.2943 [0.2817, 0.3068]	-0.2638 [-0.2774, -0.2504]
DreamSim ↓	0.6574 [0.6335, 0.6804]	0.2055 [0.1897, 0.2216]	-0.4519 [-0.4790, -0.4248]
CLIP-I ↑	0.5748 [0.5574, 0.5924]	0.8150 [0.7965, 0.8338]	0.2403 [0.2206, 0.2595]

Fixed-gray count per arm: 0. Completion improves all three feature metrics at both points; PSNR improves clearly at 10 dB, while its 4 dB interval includes zero. These results concern completion in the existing adapted receiver, including its decoder training.

S5. Additional results and processing cost

N2048. On 100 sources, the existing experiment uses two paid 1024-symbol subframes. Complete representations at 19 dB give NeST +1.806 dB PSNR over ARPC, but worse DISTS[1]: $+0.0054\,[0.0024, 0.0086]$. DISTS is an additional perceptual measure in this historical experiment, not a replacement for the four main metrics.

SNR mismatch. With both WHOLE/PARTIAL transmit policies fixed at 10 dB and RX using the true noise variance, all 300 receptions per method on 100 sources are rejected at 8 dB. All succeed at 10 and 12 dB; for each source and method the recovered prefix and reconstruction are identical at those two points. PARTIAL exceeds WHOLE by $0.660\,[0.515, 0.814]$ dB PSNR. These tested points do not locate the rejection threshold or measure the steepness of the 8–10 dB transition.

Processing cost. The following software times are from an RTX PRO 6000 with six CPU cores, batch one, 16 development images, and three measured repetitions after warm-up. They are measured processing times, distinct from the count of length checks. Ada was not timed under this protocol.

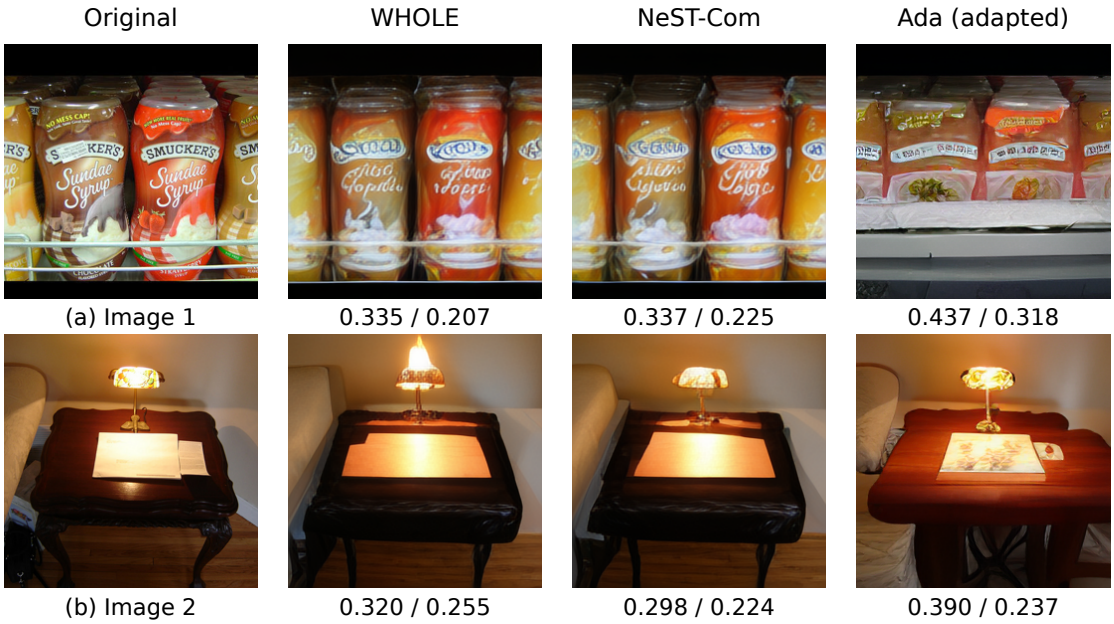
Method	SNR (dB)	TX mean (ms)	RX mean (ms)	Total mean (ms)
NeST	4	185.50	358.71	544.21
NeST	10	220.89	398.82	619.71
NeST	19	302.35	472.27	774.62
Latent JSCC	4	16.15	24.50	40.65
Latent JSCC	10	17.97	24.51	42.48
Latent JSCC	19	16.99	24.43	41.42
SwinJSCC	4	24.02	18.59	42.61
SwinJSCC	10	24.81	19.89	44.69
SwinJSCC	19	24.90	18.43	43.32
BPG + LDPC	4	2285.93	18.50	2304.43
BPG + LDPC	10	2269.87	23.15	2293.01
BPG + LDPC	19	2269.24	27.26	2296.50
ARPC + LDPC-QAM	4	716.44	719.05	1435.49
ARPC + LDPC-QAM	10	719.12	801.18	1520.30
ARPC + LDPC-QAM	19	718.19	868.69	1586.89

All underlying means, medians, and 95th percentiles remain in `inputs/timing_summary.csv`.

[1] K. Ding, K. Ma, S. Wang, and E. P. Simoncelli, "Image quality assessment: Unifying structure and texture similarity," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2567–2581, May 2022.

S6. Preserved qualitative cases and source records

The previous two-case N512 figure is retained below, with full-field images and LPIPS/DreamSim per displayed realization. Its source-ID selection history is unchanged. The main letter now uses the already displayed lamp as an editorial example, with one identical ROI across all methods; this is not a new preregistered or random selection. The crop is $x \in [80, 180)$, $y \in [0, 100)$ in the original 256-pixel grid. Main-figure PSNR/LPIPS values are for the full image, not the crop. All displayed images use N512, 10 dB, original repetition seed 6201.



The earlier N1024 figure remains in the supplemental asset directory. The native-array identifiers, source pixel hashes, wire inputs, display-image hashes, and full-precision metrics are retained in the source map and prior selection record. No image content was enhanced or repaired.

Machine-readable evidence

E1 report and complete frame/source/paired data

E2 report and complete frame/source/paired data

100-source statistics identity

N512 formal results and policies

Historical mismatch: index-only bootstrap reuse clarification

Timing data